

1 · Input

Molecular formula + literature
peak lists (IR, ^1H , ^{13}C)

blind peak lists (not traces)

2 · Generation

LLM proposes ranked
candidate structures

true enters pool 249/500 (49.8%)

3 · Verification

Self-rank / optional
forward-verify re-rank

selects true 227/249 (91%) when present

top-1 = generation recall × verification precision|recall

45.4% $\approx$ 49.8% × 91.2%

n=500 self-rank: 227/500 = 45.4%; precision|recall 227/249 = 91.2%

249 recalled (49.8%)

227

22

251

top-1

not top-1

never proposed

No re-ranking repairs the 251 never proposed

Generation decomposition on n=500 — propose is the wall, not verify